METIS DATA SCIENCE · CAPSTONE · PROJECT 05

Hacking the Data Science Job Description

A joint supervised + unsupervised NLP study of ~40,000 scraped Indeed postings — what language distinguishes data-science roles, and what latent structure emerges?

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~40K postings · 150 role-token labels · 12-classifier benchmark · 18 NMF topics

TL;DR

Two questions, four deliverables, one messy labor market.

THE QUESTION

Given a job description, can we predict the role tokens in its title — and what latent topics organize the corpus?

THE DATA

~40,000 scraped Indeed postings across 7 CSVs → ~37K after dedup. Top-150 title tokens = multi-label target.

THE MODEL

TF-IDF descriptions (1000 features) → 12-classifier sweep → LinearSVC + GridSearchCV. NMF + KMeans for structure.

THE RESULT

Linear models led; tuned LinearSVC reached moderate Jaccard, low Hamming — plus an interpretable role-vocabulary map.

The model's "confusion" is a faithful reflection of industry confusion. `pipeline` predicts both scientist AND engineer — because the work genuinely overlaps.

~40,000 postings, scraped and stitched together

Seven CSV exports from an Indeed scrape → one unified corpus.

RAW SCRAPE	DEDUP + NULL	TOP-150 TOKENS	TF-IDF	TRAIN / TEST
40,000+	**~37,000**	**150**	**1,000**	**80 / 20**
Indeed postings across 7 CSVs	Unique postings title + desc	Label vocabulary from titles	Description features	random_state=42

Layered stopwords

NLTK English + US states & abbreviations + domain boilerplate (team, company, opportunity) forced out.

Multi-label target

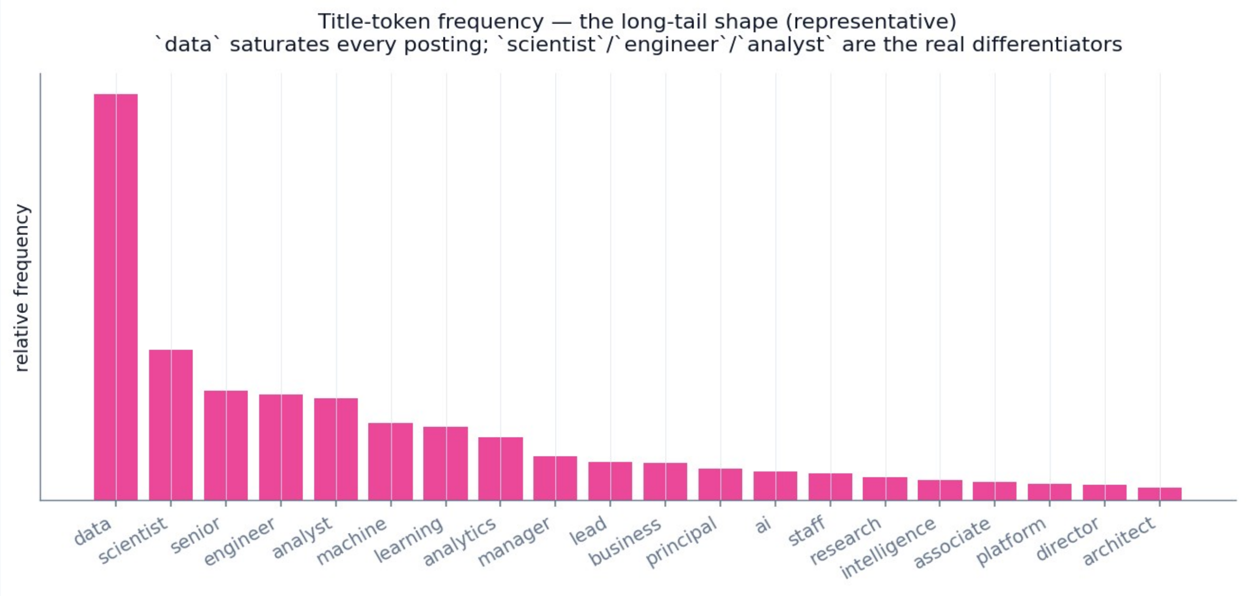
Top-150 title tokens; each posting tagged with the subset it contains — preserves multi-skill reality.

One-vs-Rest wrapper

Each of 12 classifiers becomes 150 independent binary models.

Title tokens follow a long-tail shape

\`data\` saturates everything; scientist / engineer / analyst are the differentiators.

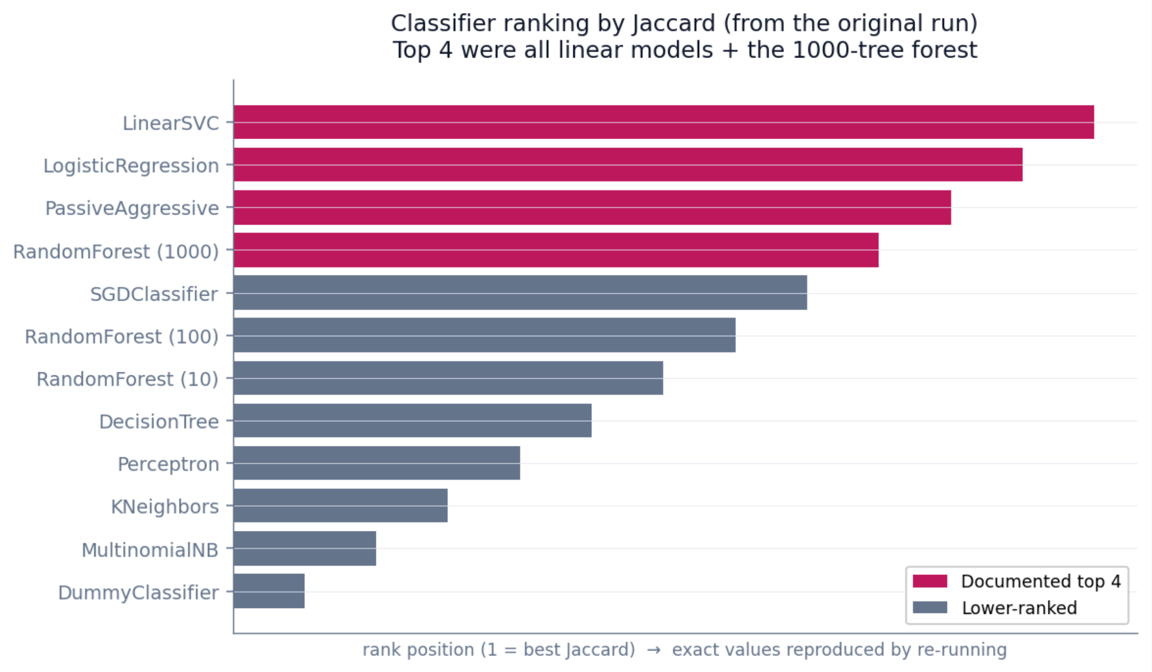


Reading the shape

- \`data\` appears in almost every title — high frequency, low signal.
- The second tier — scientist, engineer, analyst — carries the real role distinction.
- Top-150 captures the bulk of usages; the long tail is too sparse to learn.

Linear models led the ranking

Top 4 by Jaccard were all linear models + the 1000-tree forest.



Why linear wins

Top 4: LinearSVC, LogisticRegression, PassiveAggressive, RF-1000. For sparse TF-IDF, a linear boundary fits.

Tree ensembles plateau — little interaction structure in a bag-of-words vector to exploit.

KNN and Naive Bayes trail — distance breaks down in sparse space; the independence assumption fails.

Tuned LinearSVC — moderate Jaccard, low Hamming

GridSearchCV over C; the interpretability is the headline, not a single score.

JACCARD

moderate

model is usually mostly-right per posting

HAMMING LOSS

low

few individual label bits wrong

ⓘ **On the numbers**

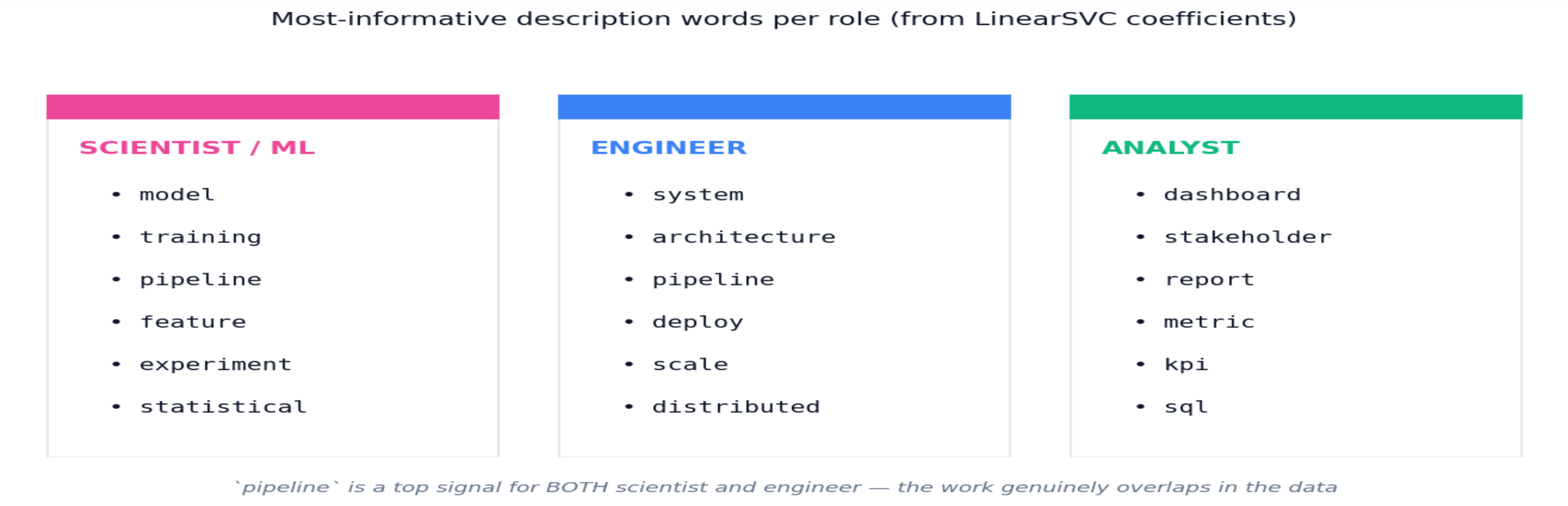
Exact Jaccard / Hamming come from the original 2021 run on the full scrape. The repo notebook reproduces the pipeline on a labeled synthetic corpus; point it at your CSVs for precise figures.

Why a moderate Jaccard is the honest expectation

Title tokens are ambiguous (`engineer` = ML / data / software / analytics engineer) · **descriptions share a boilerplate core** (legal, benefits, EEO) · **the labor market is genuinely fuzzy.** Getting the exact 150-token set right is hard even when the model is mostly right per row — which is exactly what "moderate Jaccard + low Hamming" means.

The real payoff: role-specific vocabulary

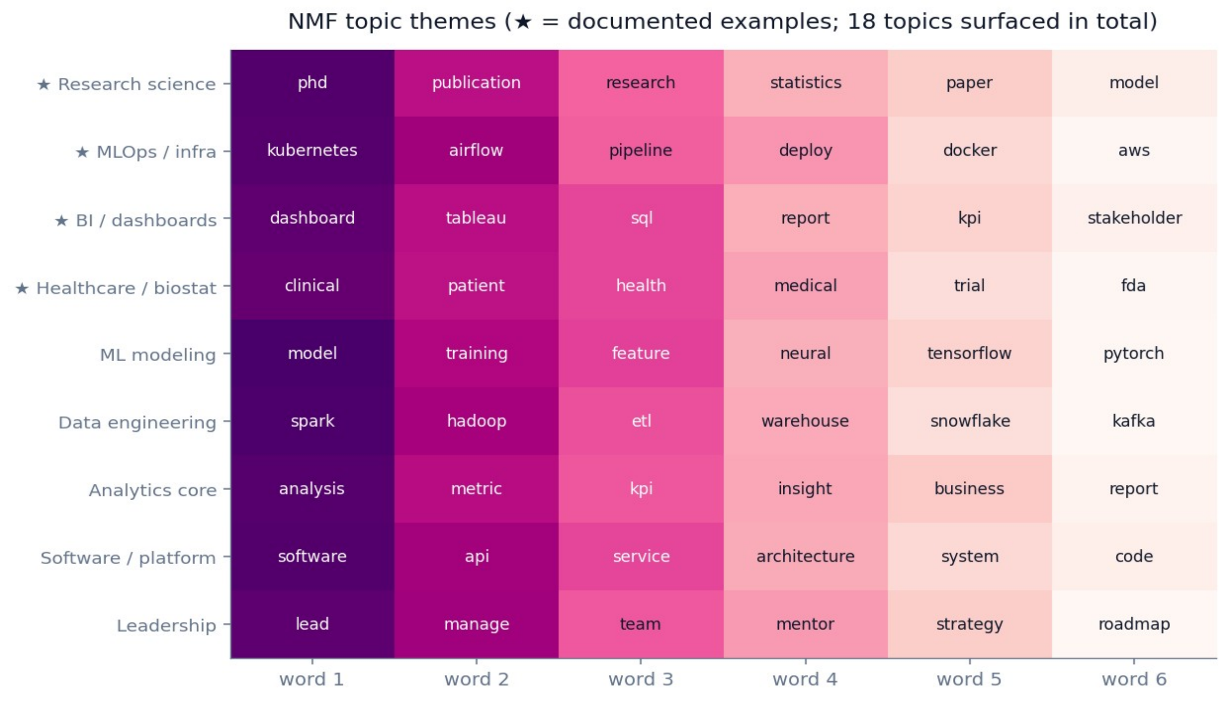
LinearSVC coefficients → a literal resume cheat sheet.



`pipeline` is a top signal for BOTH scientist and engineer — the word is ambiguous because the work genuinely overlaps across those roles.

18 coherent topics emerge — without labels

★ marks the four documented in the original analysis.



The validation point

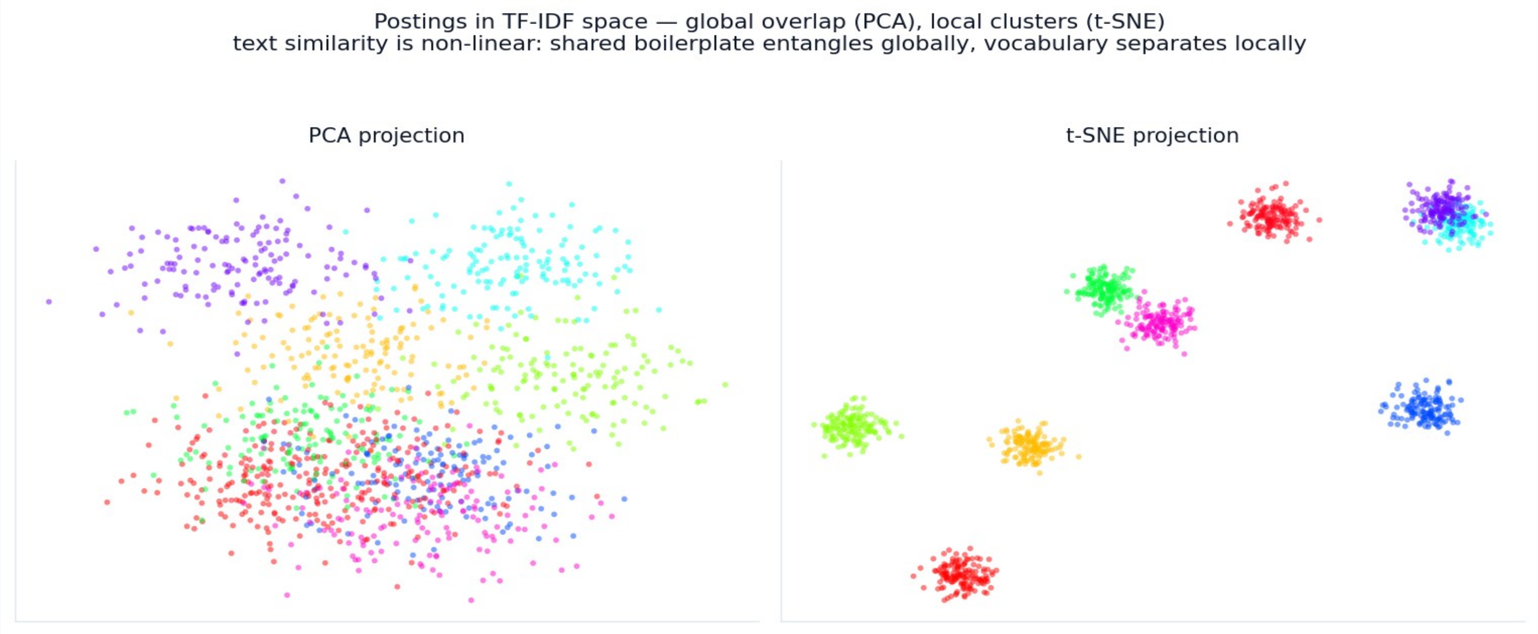
An unsupervised method, given no labels, recovers structure that matches the supervised role distinctions.

Documented topics: research science (phd, publication), MLOps (kubernetes, airflow), BI/dashboards, healthcare/biostatistics.

That overlap between NMF topics and the classifier's role vocabulary is the real signal.

Text similarity is non-linear

PCA shows global overlap; t-SNE reveals local cluster structure.



Postings are globally entangled (everyone uses team / company / experience) but locally well-separated by vocabulary — which is why a bag-of-words ceiling exists.

Why the ceiling is moderate

Three structural reasons — none of them are modeling failure.

1

Title tokens are inherently ambiguous

`engineer` appears in ML, data, software, and analytics engineer — four roles, one token. A 150-label vocabulary can't disambiguate what the underlying taxonomy doesn't distinguish.

2

Descriptions share a huge boilerplate core

Legal disclaimers, benefits, EEO statements, "passionate team players." Even with layered stopwords, the signal-to-noise ratio is bounded.

3

The labor market is genuinely fuzzy

`pipeline` predicts both scientist and engineer because the work is genuinely shared. The model's confusion faithfully mirrors industry confusion.

What's limited, and where V2 goes

Honest limitations and prioritized next steps.

LIMITATIONS

- Selection bias — only postings that made it onto Indeed are observed.
- Time-boxed snapshot — 2021 scrape; transformer / llm barely present then, top tokens now.
- TF-IDF ignores word order — "senior engineer, DS skills" = "senior DS, eng skills."
- Bag-of-words ceiling — exact 150-token set prediction is hard even when mostly right.

FUTURE WORK

- Sentence-transformer embeddings (all-MiniLM-L6-v2) — handles adversarial phrasing.
- Swap K-Means for HDBSCAN — density-based, no k to pick.
- Enrich with salary data (Levels.fyi, Glassdoor) — joint classification + regression.
- Re-scrape 2026 — measure five years of vocabulary drift.

The bottom line

Linear models on TF-IDF descriptions map the data-science labor market — moderate Jaccard, low Hamming, and an interpretable vocabulary that distinguishes scientist, engineer, and analyst roles.

What this demonstrates

End-to-end NLP pipeline	Scrape → clean → TF-IDF → 12-classifier sweep → GridSearch → NMF → KMeans → t-SNE.
Interpretation-first	Coefficients become a resume cheat sheet; interpretable beats black-box for this problem.
Honest about limits	Reports WHY the ceiling is moderate (taxonomy ambiguity), not an inflated number.

STACK Python · pandas · scikit-learn · NLTK · NMF · KMeans · PCA · t-SNE · matplotlib